

DGF-Bench

Rule Application and Evidence Reliability
in Synthetic Governance Reviews

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Abstract

Can tool-using language models apply explicit governance rules and preserve evidence through sequential project reviews? DGF-Bench contains 300 synthetic projects with documents, executable policies, authoritative structured facts, and scored review contracts. Across 899 evaluable model-project runs, strict gate success is 94.98% for Gemini 3.8 Flash, 83.29% for GPT-5.6 Luna, and 74.18% for DeepSeek v4.1 Flash; complete-route success is 76.92%, 42.33%, and 24.67%. A deterministic control executes the supplied rules and succeeds on all 1,700 gates: this information condition establishes no incremental language-model advantage. We audit all 690 gates with failed evidence components. A post-hoc lexical-or-structural provenance criterion, which tolerates field order but preserves other requirements, yields gate success of 99.06%, 85.53%, and 77.24%. These sensitivity results are not semantic adjudication. Decision confusion matrices and procurement diagnostics distinguish representation defects from substantive errors. A follow-up completes 135 runs on 15 existing dossiers; only nine, three, and zero dossiers, respectively, pass all three trajectories. An offline counterexample produces identical text in 26 Word documents despite different required decisions, showing why removing structured facts requires an information-sufficiency check. Total recorded API expenditure is USD 99.58. We release sources, documents, traces, unchanged primary scores, and analyses. The results characterize scaffolded rule application and evidence reliability; they motivate matched-input extraction studies rather than establish workforce effects.

Keywords: governance reviews; AI agents; rule application; evidence attribution; benchmark; repeatability.

1 Introduction

Enterprise project reviews transform evidence into a decision, findings, and required actions. A legal review checks contractual prerequisites; architecture and security reviews examine technical choices and protections. Software can implement a review when its inputs, policy, authority, and output contract are explicit. The relevant empirical questions are which parts require a language model and whether the resulting decisions remain grounded in observed evidence.

DGF-Bench evaluates three tool-using model endpoints on connected synthetic governance reviews. The information condition is deliberately favorable: agents can inspect executable policies and authoritative structured facts as well as generated documents. This separates execution of a specified contract from discovery of tacit rules or extraction of every premise from prose. A deterministic comparator succeeds on the entire measured task. Consequently, the contribution is an inspectable measurement of rule application, evidence conformity, and repeatability under that condition, rather than an advantage over conventional rule execution.

We report 899 original model-project runs on 300 projects, a 300-project deterministic control,

an exhaustive structural audit of the 690 gates with failed evidence components across all three models, complete decision confusion matrices, and procurement diagnostics. Another 135 runs repeat 15 existing dossiers three times per model. An offline document counterexample establishes an information-sufficiency requirement for subsequent extraction experiments. Primary scores, post-hoc sensitivity measures, repeated-run outcomes, and unexecuted protocols remain distinct.

1.1 Related work

Rule and policy application are established evaluation objects. RuLES tests adherence to explicit conversational rules with programmatic evaluation (Mu et al., 2023); LegalBench assembles expert-designed legal reasoning tasks (Guha et al., 2023). Business-process compliance monitoring predates language-model agents and provides a basis for conventional rule-based comparators (Ly et al., 2015). DGF-Bench adds connected project-review outputs and evidence contracts, but its synthetic rules are not an independently validated professional standard.

Evidence attribution is a separate issue from a correct answer. ALCE evaluates text generation with citations (Gao et al., 2023); the AIS framework specifies attribution to identified sources (Rashkin et al., 2023). Our lexical checker is narrower: it checks required observed references and excerpts. Structural matching removes some serialization sensitivity but does not establish full entailment or premise coverage. This distinction explains why a high decision score can coexist with a substantially lower strict route score.

WorkArena evaluates enterprise software interactions (Drouin et al., 2024); ITBench covers operational IT scenarios across SRE, compliance/security, and FinOps (Jha et al., 2025). τ -bench evaluates policy-guided tool/user interactions, final database state, and repeated reliability (Yao et al., 2024). CI-Work addresses contextual integrity in enterprise information flows (Fu et al., 2026), while AgentDojo evaluates adversarial tool-use settings (Debenedetti et al., 2024). DGF-Bench scores proposed reviews rather than completed operational remediation; it has no live user conversation or adversarial robustness condition. Their headline percentages are therefore not comparable. The importance of simple baselines, costs, and held-out evaluation follows the analysis of Kapoor et al. (2024).

Agentic business process management raises broader questions of process-level integration, monitoring, and escalation (Calvanese et al., 2026). DGF-Bench supplies a bounded review-workflow measurement within that agenda. Its released traces support studying where a model’s contribution might lie before adding it to an otherwise sufficient rule engine.

1.2 Research boundary and companion manuscript

This empirical study shares its original September 2026 experiment with the same author’s *The Last Human Gate: Forward Deployed Engineering and the Automation of Enterprise Governance*. The control, saved-trace audit, and repeated trajectories supplement that experiment; they are not an independent replication on new projects. The companion retains the broader FDE argument, labor accounting, and workforce scenarios. Here those implications are a short discussion, not the principal empirical claim.

The workflows are author-designed and informed by practitioner experience. Their definitions, artifacts, public rules, and scoring interface are released. Inspectability does not establish that this policy represents any particular enterprise. The generator and evaluator share rule definitions, so their agreement is internal consistency rather than independent validation.

2 DGF-Bench: measured review performance on 300 synthetic projects

2.1 Question, population, and information available to the agents

This experiment asks whether an AI agent can apply a declared governance policy, identify the required findings and actions, cite observed evidence, respect its authority, and carry a review through a connected route. It measures review decisions and proposed actions, not implementation of remediation or changes in human workload. The field-validation studies proposed later in this paper have not been conducted.

Three practical questions organize the analysis: (1) does the model reach the disposition required by the rules, (2) does it produce all the supporting work needed for a valid review, and (3) does that reliability survive a sequence of reviews at an acceptable API cost? The unit called a *case* or *dossier* is one generated project; a *gate* is one checkpoint on its route. Three models reviewing the same 300 projects produces 900 planned model-case runs, not 900 independent business projects.

The dataset contains 300 fictional projects: 100 Buy, 100 Integrate, and 100 Build cases. Buy and Integrate contain six gate occurrences each, and Build contains five, for 1,700 occurrences per model and 5,100 planned observations across three models. Cases were generated before these evaluations at difficulty setting 4, with sampling conditioned to improve decision coverage. Difficulty 4 is a generator configuration, not a validated scale of cognitive difficulty. For example, before decision-balancing rejection sampling, the architecture IP-overlap draw has probability $0.08 + 0.03 \times 4 = 0.20$ and the shared-service-principal draw has probability $0.10 + 0.03 \times 4 = 0.22$. Supplier generation permits sanctions hits and up to two failed mandatory criteria, and may select a non-optimal offer. Subsequent balanced sampling changes accepted outcome frequencies; these generator probabilities are not observed failure rates. The executable frozen source defines the setting, and no difficulty-level ablation is reported.

Project context, owners, budgets, data classifications, technical architectures, documents, and simulated system records derive from canonical case facts. Documents can be missing, stale, or inconsistent. Specialist reviews precede a final General review; the routes are examples rather than a universal enterprise standard.

The agent receives gate objectives, a structured output contract, an explicit decision policy, and tools to read allowed evidence and query the synthetic environment. In particular, the policy includes executable rule definitions and identifies an authoritative structured `REVIEW_FACTS` snapshot. The evaluator-only reference file is inaccessible through the agent’s tools. This is nevertheless a strongly scaffolded test of policy application: the agent need not discover the rules or reconstruct every decision-relevant fact from unstructured documents. Architecture images are available under the vision setting; image availability is not evidence that image interpretation was necessary for success. These design choices limit claims about expert reasoning on independently collected enterprise dossiers.

The dataset audit reports 300 distinct project identifiers, architecture signatures, and complete decision-fact signatures, but only 267 distinct whole-case finding patterns. Gate-specific relevant fact signatures include 142 of 200 Compliance cases and 241 of 300 Security cases. All rules applicable to the selected phases are covered according to the generator’s audit. Phase-adjusted normalized decision entropy ranges from 0.9971 to 1.0000. Balanced labels improve coverage, but do not estimate enterprise prevalence or establish independent business validity.

2.2 What is generated, and what the model actually does

The case generator is a program using seeded facts, rules, and document templates. The AI model under evaluation is the reviewer, not the author of the fictional need or architecture. A Buy scenario represents acquiring a solution; Integrate represents connecting existing systems; Build

represents developing a new project. Those scenarios are given budgets, users, owners, dates, data classifications, and technical constraints. A generated name is a scenario label, not an independently elicited customer requirement.

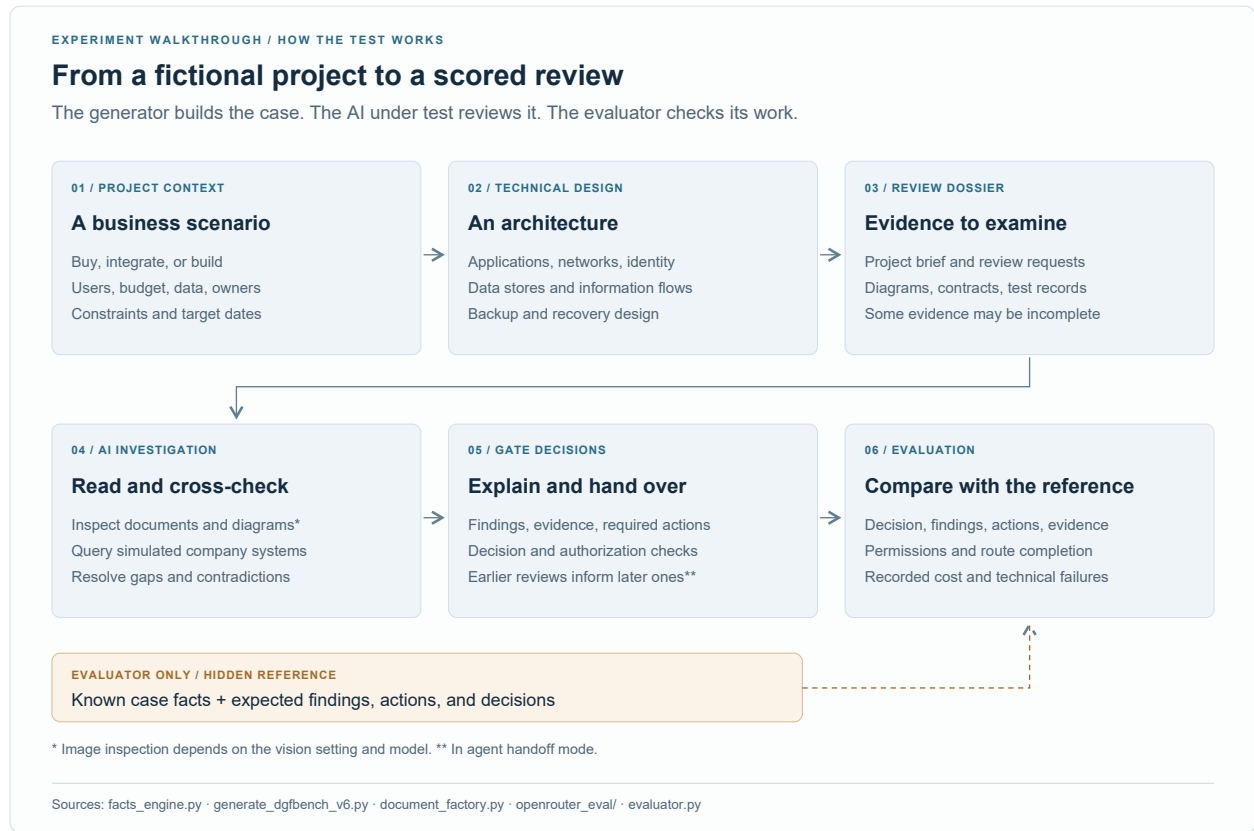


Figure 1: Experimental workflow. The generator constructs the project and evidence; the tested model investigates and submits reviews; the evaluator uses a separate reference. The diagram describes the implemented test, not a measured enterprise deployment.

1. **Construct the project facts.** The program fixes what is true in the synthetic case, such as the requested budget, whether a supplier has supplied a document, or whether a restore test succeeded. Decision-coverage sampling produces cases with different required outcomes.
2. **Generate the design and dossier.** The same case drives an architecture diagram, project charter, gate review requests, and evidence files. A reviewer may receive a technical design, network/data-flow tables, contract records, security evidence, or readiness tests, depending on the gate (Table 1). Some presented evidence is incomplete or stale.
3. **Start a gate review.** The harness supplies the project context, gate and phase, applicable policy, permitted evidence and tools, and the answer schema. The model decides which allowed records to inspect. It can list and read evidence, query simulated systems, request information, and inspect existing authorization mandates.
4. **Submit a decision.** The model returns a disposition, finding identifiers, proposed actions, evidence references and supporting excerpts, an authorization indicator, and a rationale. A typed submission is recorded. Producing a syntactically valid record alone does not satisfy the substantive score.
5. **Continue the route.** In the tested **agent** mode, the harness passes actual upstream model reviews to later checkpoints. The final General review consolidates specialist opinions and its own checks. The tested routes are listed in Table 2.
6. **Evaluate and retain the trace.** A deterministic evaluator compares the submitted record

and observed tool events with the reference. It records component scores, full gate and route success, critical misses, false approvals, provider failures, and usage. It is not another LLM judge. Failed or incorrect decisions are retained; infrastructure exclusions are reported.

Table 1: Concrete examples of generated review material. The contents vary with the route and gate. Authoritative structured snapshots are available in addition to these artifacts.

Material	Examples in the case	Review question
Project context	Charter and review request (Word), project context (JSON)	What is proposed, for whom, at what cost, and under whose ownership?
Architecture	Diagram (SVG/PNG), technical design, network and flow tables	Do the design facts meet the declared constraints?
Security	Identity, vulnerability, monitoring, and resource records	Are the protections required by the policy present?
Supplier and legal	Vendor evidence, contract versions, data-processing agreement status	Are the supplier and contractual prerequisites satisfied?
Readiness	Backup, restore/failover and load-test records; runbook	Can the proposal demonstrate the required operational readiness?

Table 2: The three routes actually evaluated; each project ends with General. These are the complete route sequences evaluated in this study.

Route	Cases	Ordered gates
Buy	100	Procurement, Legal, Compliance, Security, IT, General
Integrate	100	IT, Architecture, Security, Legal, Compliance, General
Build	100	IT, Architecture, Security, Tech Readiness, General

The five dispositions are **GO** (proceed), **GO_WITH_RESERVATIONS** (proceed under conditions), **REWORK** (revise the proposal), **SUSPENSION** (wait for a prerequisite), and **NO_GO** (do not proceed as submitted). Approval is therefore not the target outcome for every dossier: a justified refusal is a successful review. Tool actions take place in the synthetic environment; a request to fix a defect is not evidence that the defect was fixed.

2.3 A generated project and a recorded successful review

Case DGF-BLD-035200_build, *Project Falcon—Secure Vendor Portal*, provides a concrete example. Its generated HR context has 1,000 users, a requested budget of EUR 500,000, confidential data, high business criticality, and a primary region of UAE North. Figure 2 reproduces its actual evidence diagram. It depicts identity, application and data services, monitoring and backup, and a secondary recovery region. Its labels are claims in the dossier to be checked, not an independently validated production blueprint.

In Gemini’s recorded Tech Readiness review, the authoritative record has a draft runbook. The agent identifies TR-OPS-001 and proposes completing the operational handover. Other required readiness checks pass. It cites the observed excerpt "**runbook_status**": "**draft**", consults the active mandate, and executes the simulated conditional-approval tool with a condition to approve the runbook before production cutover. The tool reports **EXECUTED** and explicitly leaves the finding unresolved. The submitted outcome is therefore **GO_WITH_RESERVATIONS**, with the finding and required action retained and the authorization indicator set to true.

All three models pass all five gates of this particular dossier. Their observed dispositions are **GO**, **GO**, **GO**, **GO_WITH_RESERVATIONS**, and **GO** in route order. This example was selected after evaluation to explain the mechanics, not as a representative sample or an additional experiment. The study-level counts include every evaluable case, including unsuccessful reviews. The public research directory includes `worked_examples.json`, with the original archive-relative paths and selected recorded

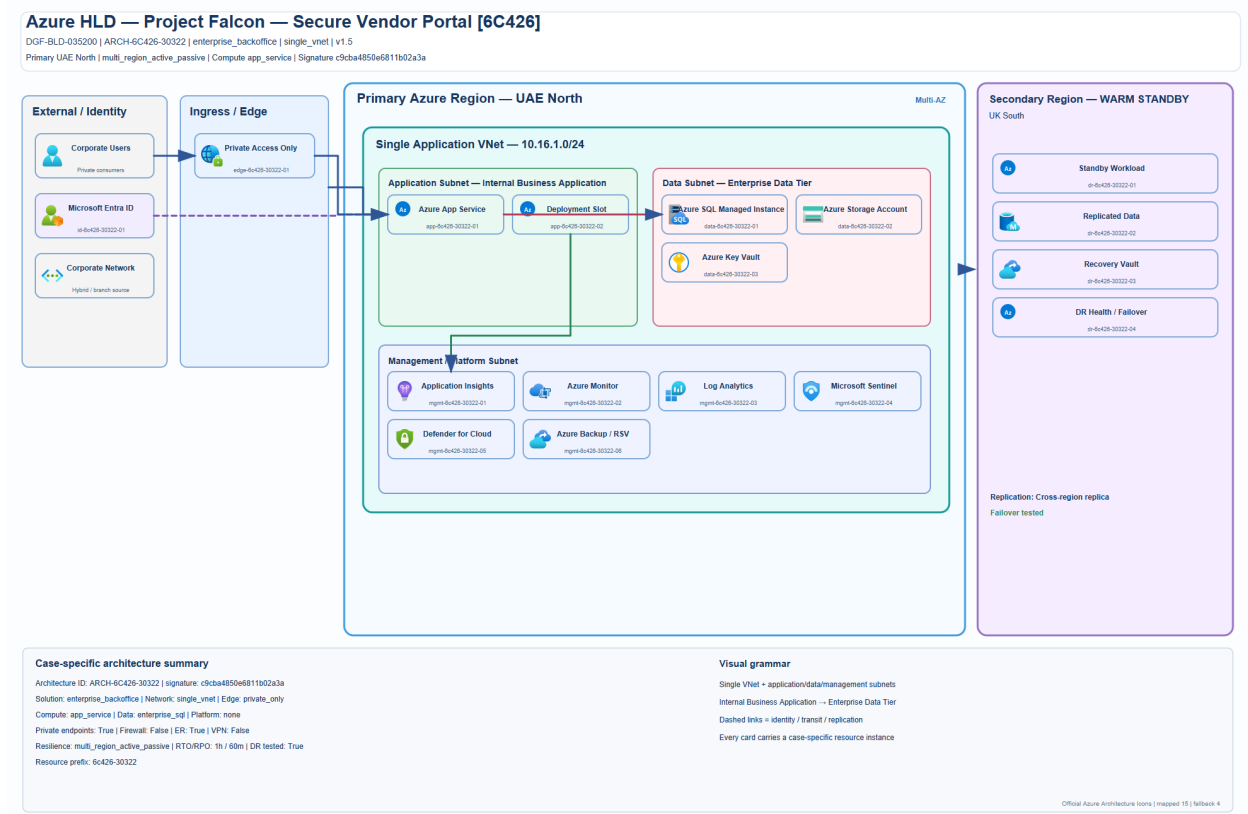


Figure 2: An actual generated architecture document from Project Falcon. Full-resolution evidence is in the released dataset. The figure illustrates what a dossier can contain; it does not imply that this image was necessary to obtain the correct review outcome.

outputs for this example and the evidence-failure example below. The complete traces remain in the release.

2.4 Protocol, completion, and outcomes

Table 3 identifies the three tested endpoints. Each receives the same case population, policy, tool interface, and task limits; no model is trained or fine-tuned by this experiment. The comparison concerns these deployed endpoints under the recorded conditions, not different implementations of a human review process. Configuration was temperature 0, vision **auto**, agent-produced handoffs, at most 20 turns and 40 tool calls per gate, and 8,192 output tokens per turn. Gates remained sequential within a project. The last resume used 12 workers, four per model, and a USD 110 cost cap; an earlier invocation used six workers and a USD 65 cap. Data collection spanned 22–23 September 2026. Resumption retained completed cases and valid gate checkpoints. Historical provider/key-limit failures remain in the released traces. The 78.73-minute elapsed time in the last run statistics describes that resume only.

Of 900 planned model–case runs, 899 are evaluable: 300 for DeepSeek, 299 for Gemini, and 300 for Luna. Gemini case DGF-INT-035128_integrate failed at its first IT gate with a provider generation error (**finish_reason=error**); it is excluded as an infrastructure failure. Thus 5,094 of 5,100 planned gates are evaluable, with all expected gates attempted in the evaluable cases. There are no recorded final agent-protocol failures or model-identifier mismatches. Provider-reported identity is not an independent audit of underlying weights. DeepSeek was served by multiple OpenRouter providers; Gemini and Luna were recorded through Google AI Studio and OpenAI.

Scoring uses DGF-decision-v8.2-authorized-review. A gate succeeds strictly only if its disposition, finding set, action set, evidence provenance, and authorization indicator all satisfy the

Table 3: Models used in the experiment. Names are short labels for the exact requested OpenRouter identifiers, not claims about independently verified model weights.

Model label	Requested identifier	Recorded serving provider
DeepSeek v4.1 Flash	deepseek/deepseek-v4.1-flash	Multiple providers
Gemini 3.8 Flash	google/gemini-3.8-flash	Google AI Studio
GPT-5.6 Luna	openai/gpt-5.6-luna	OpenAI

reference. Findings and actions require F1 equal to one. Evidence requires an observed reference and exact excerpts supporting the relevant field/value tokens. This is lexical provenance checking, not independent verification of arbitrary reasoning. Authorized conditional approvals can change the effective reference disposition; the evaluator verifies the mandate and recorded action. General combines effective reference opinions, so an upstream error can also cause a downstream failure. Complete-route success requires every gate to succeed strictly; it does not mean that remediation was executed.

For model m , let s_{mcg} be one when gate g of case c passes every component, and zero otherwise. The two principal rates are

$$\text{GateCSR}_m = \frac{\sum_c \sum_{g \in G_c} s_{mcg}}{\sum_c |G_c|}, \quad \text{RouteSuccess}_m = \frac{\sum_c \prod_{g \in G_c} s_{mcg}}{|C_m|},$$

where C_m contains the evaluable cases and G_c their expected gates. This gate rate is weighted by gate count: a six-gate route contributes six observations and a five-gate route contributes five. A model passing four gates of a five-gate case has 80% gate success for that case but zero complete-route success. Decision-only accuracy checks just the submitted disposition. A critical miss is an omitted reference finding classified as critical; a false approval is GO or GO_WITH_RESERVATIONS when the effective reference requires rework, suspension, or refusal.

Gate confidence intervals use a route-stratified bootstrap of whole cases, preserving dependence between a project’s gates (2,000 draws, seed 81931; a case-count Wilson fallback handles degenerate intervals). Route intervals use Wilson bounds. Pairwise differences use the same 299 cases for all models, resampled together within routes (10,000 draws, NumPy seed 81931). These are descriptive intervals conditional on the synthetic sampling design. No preregistered analysis or independently adjudicated ground truth is claimed, and one trajectory per model and case does not estimate run-to-run inference variability.

2.5 Overall and route-level results

Table 4: Measured synthetic benchmark outcomes. Gate CSR requires complete conformity; route success requires all gates. Cost includes recorded failed attempts.

Model	Cases	Gate CSR [95% CI]	Route success [95% CI]	USD
DeepSeek	300	74.18% [71.88, 76.47]	24.67% [20.13, 29.84]	10.09
Gemini	299	94.98% [93.86, 96.04]	76.92% [71.82, 81.34]	71.88
Luna	300	83.29% [81.29, 85.24]	42.33% [36.87, 47.99]	5.05

Gemini succeeds strictly on 1,609 of 1,694 gates, Luna on 1,416 of 1,700, and DeepSeek on 1,261 of 1,700. Corresponding complete-route counts are 230 of 299, 127 of 300, and 74 of 300 (Table 4). Mean weighted component scores, which are not strict success rates, are 99.68%, 97.67%, and 95.02%, respectively. The distinction matters: strong average components can coexist with many incomplete project reviews.

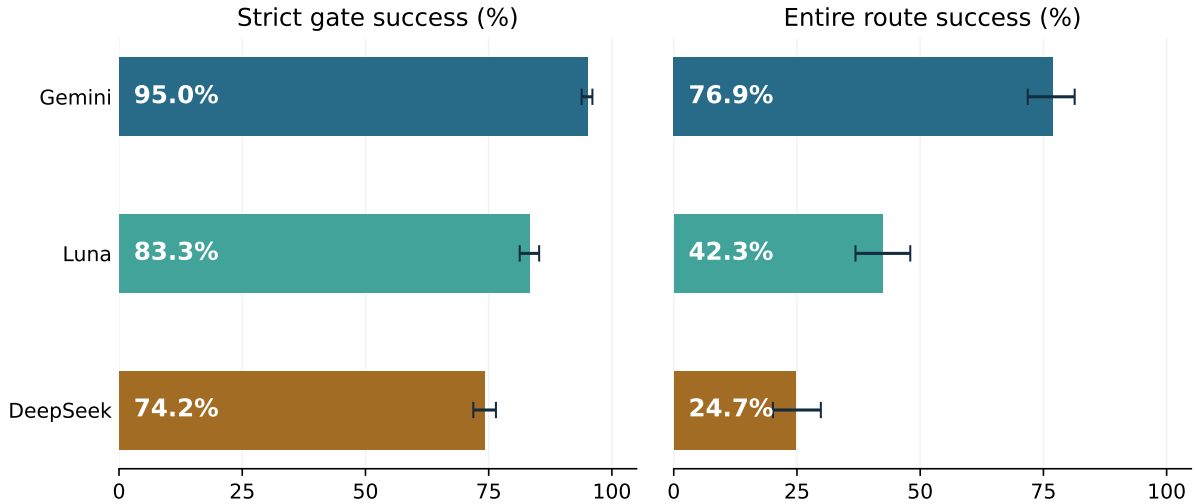


Figure 3: Measured gate and complete-route review outcomes. Intervals are computed over independent case units rather than treating all gates as independent trials. These are model observations on synthetic dossiers, separate from the workforce scenario figures.

On the 299 common cases, CSR is 94.98% for Gemini, 83.23% for Luna, and 74.09% for DeepSeek. Paired differences are 20.90 percentage points for Gemini minus DeepSeek (95% interval 18.30–23.44), 11.75 for Gemini minus Luna (9.62–13.93), and 9.15 for Luna minus DeepSeek (6.67–11.63). The ranking persists on the matched cohort; this is not a universal model ranking.

Table 5: CSR and complete-route counts by project route. Gemini excludes one Integrate case.

Route	DeepSeek: CSR; routes	Gemini: CSR; routes	Luna: CSR; routes
Buy	67.83%; 13/100	97.67%; 87/100	74.67%; 20/100
Integrate	78.17%; 32/100	95.12%; 75/99	87.67%; 52/100
Build	77.00%; 29/100	91.60%; 68/100	88.40%; 55/100

Gemini’s route success falls from 87 of 100 Buy cases to 68 of 100 Build cases. Luna succeeds on 20 Buy, 52 Integrate, and 55 Build cases; DeepSeek succeeds on 13, 32, and 29, respectively (Table 5). These differences do not isolate the causal effect of route length, because gate composition and case content also differ.

Table 6: Strict success by gate family (percent). Counts vary by route composition; complete denominators and intervals are supplied in the released tables.

Gate	DeepSeek	Gemini	Luna
Architecture	75.00	85.43	89.50
Compliance	88.00	100.00	96.00
General	68.67	87.96	61.67
IT	87.00	100.00	97.67
Legal	68.50	99.50	80.50
Procurement	43.00	96.00	33.00
Security	72.33	98.66	92.00
Tech Readiness	71.00	89.00	97.00

Procurement is the lowest-CSR gate for both Luna (33%) and DeepSeek (43%). Luna’s General CSR is 61.67%. Luna exceeds Gemini’s descriptive CSR in Architecture and Tech Readiness (Table 6); these local comparisons are exploratory, without multiplicity-adjusted tests or a claim of specialist superiority.

2.6 Evidence failures, unsafe approvals, and cost

Disposition accuracy alone is 100% for Gemini, 98.12% for Luna, and 95.24% for DeepSeek. All 85 Gemini strict failures concern evidence conformity; findings, actions, dispositions, and authorization indicators match the reference. Evidence is the sole failing component in 221 of 284 Luna failures and 338 of 439 DeepSeek failures. Invalid excerpts and missing or unobserved references are grouped by these diagnostics; not all are harmless formatting defects. Changing the evidence rule after observing results would change the estimand. A semantic evidence metric would require a separately declared analysis.

A recorded example of a correct decision with a failed proof. In DGF-BLD-035201_build, Gemini reads the Architecture facts: an API gateway is required but absent, and measured latency is 55 ms against a 20 ms target. It correctly names ARCH-API-001 and ARCH-PERF-001, proposes adding the gateway and remediating latency, and submits REWORK. Two attempted conditional approvals are rejected by the tool: one omits an open finding and the other includes a finding ineligible for risk acceptance. The submitted refusal to proceed is consistent with those restrictions.

The gateway excerpt reverses two fields. It places `api_gateway_present` before `api_gateway_required`; the observed record has the reverse order. The values are correct but the excerpt is not an exact observed sequence. Evidence fidelity is 0.5 for the two findings, so strict success is zero while disposition, finding, action, and authorization scores are all one. The weighted component score is 0.95. This selected trace explains exactly how a fluent, substantively correct review can fail this benchmark’s evidence contract; it does not justify silently relaxing that contract after seeing the results.

DeepSeek records one critical finding miss and one false approval, both in the Architecture review of DGF-INT-035136_integrate. An address overlap required NO_GO; at forced finalization, the model submitted GO with an empty finding set, no evidence, and a “placeholder” rationale. The response passed the submission schema but failed the substantive score. This observation combines decision quality with the ability to finalize under resource limits.

Luna records three critical misses and one false approval. In case DGF-INT-035197_integrate, its Legal review approved a dossier with a missing DPA, while its own rationale said that REWORK was required. The other critical misses concern private endpoints in Security and an upstream NO_GO in General. The latter can include error propagation and should not be interpreted as an independent new safety event. Gemini has no observed critical miss or false approval in its evaluable runs; this does not establish zero risk. Recorded truncated-response counts are 2 for DeepSeek, 14 for Gemini, and 0 for Luna, without implying that these counts equal incomplete cases.

The usage ledgers sum to USD 87.015829172, agreeing with the cumulative budget counter. Gemini accounts for USD 71.88, DeepSeek USD 10.09, and Luna USD 5.05, including recorded retries and failures. Approximate cost per evaluated dossier is USD 0.240, 0.034, and 0.017, respectively. Luna therefore combines a higher overall strict score with lower recorded cost than DeepSeek in this execution. Costs depend on provider routing, caching, price, and retry history; they are not a labor-cost comparison or a permanent pricing claim.

3 What the scores establish

3.1 A near-saturated disposition task and a stricter trace contract

Gemini’s 100% disposition accuracy is a strong result under the declared conditions. It also indicates that this sample offers little remaining discrimination on that component for this endpoint. The model can read explicit rules and authoritative structured facts; this experiment does not isolate how much ability is needed to recover those facts from documents. The deterministic control now executed on the same 300 dossiers reaches full strict success. It demonstrates that the supplied

executable rules and structured facts are sufficient to solve this instance of the task without model inference. Model success remains an observation about model-based execution, but does not establish an incremental benefit over direct rule execution.

Strict success answers a different question: does the entire submitted review satisfy the implemented contract, including exact evidence provenance? It is useful when downstream processing requires precise traceability, but it is not a direct measure of independent expert judgment. The selected Architecture example shows a correct decision and correct values with a failed excerpt because two fields were reversed. The Legal false approval by Luna and DeepSeek’s placeholder approval illustrate materially different errors. Those classes should not be collapsed into a single narrative about reasoning failure.

Table 7: Separate observed outcomes. Evidence-only failures have no other failing scored component; they are not all established to be harmless formatting errors.

Outcome	DeepSeek	Gemini	Luna	Rules
Correct dispositions / evaluable gates	1,619/1,700	1,694/1,694	1,668/1,700	1,700/1,700
Strictly successful gates	1,261	1,609	1,416	1,700
Evidence-only gate failures	338	85	221	0
Other strict gate failures	101	0	63	0
False approvals	1	0	1	0
Critical finding misses	1	0	3	0

The strict gate-route gap for Gemini is consequently a compounding of evidence-conformity failures across routes, not evidence of incorrect final dispositions. A strict route fails when any component at any gate fails. This is the intended operational definition, but the figure 76.92% must be read with that definition. We retain both the original scoring rule and the component breakdown. A declared post-hoc sensitivity analysis can quantify representation effects, but cannot replace the primary score or imply that every lexical failure is a dangerous decision.

3.2 Executed deterministic control

A separate generation process reads each dossier’s public route and gate contract. For each occurrence it reads the authoritative snapshot through the same public evidence reader, executes `rule_definition_python` from the supplied policy, combines finding dispositions using the declared priority order, and sets the authorization indicator using the declared disposition list. Finding construction and result aggregation are implemented from that public interface. The policy functions are the executable definitions supplied to the language models; they are not a separately learned policy or an independently validated governance standard.

The comparator retains the base disposition and never requests optional conditional risk acceptance, a strategy explicitly permitted by the policy. It cites complete observed JSON snapshots as exact evidence excerpts and propagates its own earlier decisions to General. This exploits the absence of a quote-length cap; the program is not subject to the models’ per-turn output-token limit. It does not attempt document extraction or image interpretation. A read guard blocks evaluator-only reference files during generation, and the generation process asserts that neither the evaluator nor scorer module was imported. All predictions and observation records are written before a second process invokes the frozen scorer, which is allowed to read the hidden references for evaluation.

The result is **1,700/1,700 strict gate successes and 300/300 complete routes**, with no false approvals or critical misses. Model API expenditure is zero; programming and local compute are not costed. On the 299 cases shared with Gemini, the control also succeeds on all 1,694 gates. Optional approval choices can differ between systems, and each is scored against the protocol’s effective reference; full conformity does not mean identical submitted decisions.

This is a scaffold-sufficiency control. It does not validate the business rules independently, because

the public policy and evaluator share their definitions. It does show that an LLM is unnecessary for achieving the measured endpoint when those functions and structured facts are already available. The main unresolved model contribution therefore lies in tasks that this run did not isolate: recovering facts, resolving ambiguity, and interacting with less structured operational evidence. Claims about such a contribution require the proposed ablations and external validation, rather than a higher score on this saturated contract.

3.3 Executed structural audit of Gemini’s 85 failures

We inspected every Gemini strict-failure gate against its saved submission and pre-submission read events. The 85 failed gates contain 84 flagged finding-support items: a gate may contain several items, while a citation-only failure may contain none. The audit parses each quote as a JSON object fragment, rejects duplicate keys, and checks all quoted field values against a single observed object while ignoring key order and numeric serialization such as 30 versus 30.0. Booleans remain distinct from numbers; 10 cannot match 100. A second diagnostic records where values exist across different objects, without treating a flattened join as a faithful single-object excerpt. This is a post-hoc structural test, not independent expert adjudication.

Table 8: Exhaustive post-hoc audit of Gemini strict failures. Categories describe recorded representation and observation defects, not new semantic pass rates.

Gate category	Gates	Item-level explanation
All flagged quotes match observed objects after structural normalization	69	75 flagged finding items preserve the checked field values within an observed object.
Flattened cross-object excerpts	9	9 flagged items combine observed values from separate objects; paths are retained for inspection.
Required evidence-tool observation absent	7	All cite <code>UPSTREAM_DECISIONS</code> without the corresponding read event.

For example, a flattened high-availability excerpt combines business criticality from the project object with redundancy flags from the architecture-profile object. Those values are present, but the quoted combined object was not observed. In the seven citation cases, upstream reviews were also included in the gate prompt; the defect is the missing required evidence-tool read, not proof that the model never received the underlying information. Neither a corrected quote nor an invented read event is inserted into the original trace.

The audit identifies representation-related causes for the 78 excerpt-failure gates and a separate observation-contract cause for seven gates. It supplies no independent judgment that every policy inference is justified or that every necessary premise was cited. The single-object matching check accepts relevant field values rather than proving full entailment, as does the original lexical checker. We retain the original 94.98% strict gate and 76.92% strict route rates and report the audit alongside them. No revised whole-benchmark semantic rate is claimed. Complete excerpts, observed contents, matched paths, categories, and source trace identifiers are released with the audit script.

3.4 All-model structural sensitivity and procurement failures

We apply the same post-hoc check to every original gate with evidence fidelity below one: 354 DeepSeek, 85 Gemini, and 251 Luna gates. A gate passes the relaxed endpoint if it originally passed strictly, or if all unchanged non-evidence components pass, its required citations were observed, and each originally unsupported finding has a relevant exact field/value subset within one observed object. Cross-object joins are not promoted. Quotes and read events are not repaired. Original lexical successes remain accepted, so this is a lexical-or-structural sensitivity endpoint, not a replacement semantic metric. A matching relevant field need not supply every premise of a finding.

Table 9: Original strict versus post-hoc lexical-or-structural success on the same saved trajectories. Relaxation changes the evidence criterion only; it is not a new model result.

Model	Gates		Complete routes	
	Strict	Relaxed	Strict	Relaxed
Gemini	1,609/1,694	1,678/1,694	230/299	284/299
Luna	1,416/1,700	1,454/1,700	127/300	144/300
DeepSeek	1,261/1,700	1,313/1,700	74/300	85/300

Gate success rises from 94.98% to 99.06% for Gemini, 83.29% to 85.53% for Luna, and 74.18% to 77.24% for DeepSeek. Complete-route success rises to 94.98%, 48.00%, and 28.33%, respectively. The large Gemini gate-route gap is therefore highly sensitive to serialization. The smaller changes for the other endpoints show that this explanation does not transfer uniformly across models.

Table 10: Classification of evidence-only failed gates. “Unresolved” means not accepted by this structural parser, not a demonstrated false claim. Categories are mutually exclusive.

Category	DeepSeek	Gemini	Luna
Same-object matches for all failed finding items	52	69	38
Cross-object excerpt remains	2	9	46
Citation/observation defect	109	7	1
Unresolved structural or support defect	175	0	136
Total evidence-only failures	338	85	221

Procurement localizes an especially important contrast. There are 100 evaluable Procurement gates per model. Gemini makes 100 correct decisions and passes 96 strictly; all four failures are budget-related cross-object excerpts. DeepSeek makes 97 correct decisions but passes only 43 strictly, with 54 evidence-only failures. Luna makes 96 correct decisions but passes 33 strictly, with 63 evidence-only failures. Structural relaxation recovers no additional DeepSeek Procurement gate and only one Luna gate. Many failed items are not parseable as object fragments; others combine separate objects or lack valid support. These categories require semantic review before being called hallucinations. Procurement is thus mainly an evidence-contract bottleneck in this sample, despite some substantive decision errors. Per-rule classifications and original quotes are included in the released item-level audit.

Independent semantic adjudication remains unperformed. A suitable rubric would separately label observed access, source/span fidelity, factual entailment, and sufficiency for the policy finding; two masked domain-qualified adjudicators would label independently and resolve disputes under a fixed procedure. Both failures and a probability sample of passes should be assessed. A full semantic route rate requires adjudicating all required support on the routes counted, not just forgiving selected failed quotes.

3.5 Decision confusion and the effective reference

The full confusion matrices in Appendix A show 81 incorrect DeepSeek dispositions and 32 incorrect Luna dispositions; Gemini has none under the effective-reference score. DeepSeek’s errors are mostly more restrictive than the reference under the policy’s priority order (76/81), including 35 GO-to-REWORK and 23 GO_WITH_RESERVATIONS-to-REWORK outcomes. Luna has 10 more-restrictive and 22 less-restrictive errors, including 17 SUSPENSION-to-REWORK outcomes. A less-restrictive error between blocking decisions is not automatically a false approval; each of these two models has one scored false approval.

The effective reference incorporates verified conditional approvals and their authorized upstream consequences. It is not always the initial label. Agreement with the unmodified base reference is

1,553/1,700 for DeepSeek, 1,553/1,694 for Gemini, and 1,568/1,700 for Luna, versus 1,619/1,700, 1,694/1,694, and 1,668/1,700 with the effective reference. Gemini’s 141 base-label differences comprise 139 local conditional approvals and two General outcomes reflecting upstream authorized changes. The evaluator checks mandate and approval conditions; the model cannot make a decision correct merely by declaring it approved. Reporting both agreements clarifies the target: the primary metric measures conformity after allowed actions, while initial-label agreement measures a different, action-insensitive target.

3.6 Executed repeated trajectories on 15 dossiers

Before launching follow-up calls, we selected five dossiers without replacement per route from sorted identifiers with Python `random.Random(23092026)`. Selection did not use model outcomes. The local plan fixed three new trajectories per model–case pair, giving 15 dossiers, three endpoints, 135 runs, and 765 gates. This is a local analysis plan, not an external preregistration. The original historical trajectories are not counted as a fourth exchangeable repeat and are not pooled into the follow-up rates.

Collection on 23–24 September 2026 used byte-verified copies of the existing dossiers and the frozen benchmark source. All three repetitions have identical recorded protocol identities: temperature zero, agent handoffs, automatic vision, 20 turns, 40 tool calls, and 8,192 output tokens per turn. There were three workers, one per model. Each repeat had a separate output directory. Infrastructure recovery retained completed gate prefixes within that trajectory; successful completed routes were not rerun or selected on their score.

All 135 final runs are evaluable and all 765 gates attempted. Retained error checkpoints record 46 key-limit refusals and two provider-finish errors. Two Gemini runs resumed seven completed gates in total. One truncated response is recorded for Gemini and no resolved-model mismatches are recorded. The total USD 12.5598870228 includes unsuccessful paid attempts; all ledger entries have known costs. The authorization was increased from USD 20 to USD 50 after a separate API-key limit interrupted collection. This did not change the case sample, repeat count, or inference settings.

Table 11: Three fresh trajectories on the same 15 dossiers. Strict gate counts are out of 85 per repeat; complete-route counts are out of 15. The last column requires all gates to pass on all three trajectories of a dossier.

Model	Strict gates / 85			Complete routes / 15			All three / 15
	R1	R2	R3	R1	R2	R3	
Gemini	82	82	81	12	12	11	9
Luna	73	71	67	8	7	4	3
DeepSeek	63	64	60	3	4	4	0

Table 12: Follow-up rates pooled over three trajectories, with exploratory 95% case-cluster bootstrap intervals in brackets. Every model has 255 scored gates and 45 completed runs.

Model	Strict gate success	Complete-route success	API cost (USD)
Gemini	96.08% [93.33, 98.43]	77.78% [62.22, 91.11]	10.3889
Luna	82.75% [75.29, 89.80]	42.22% [22.22, 62.22]	0.7499
DeepSeek	73.33% [66.67, 79.61]	24.44% [13.33, 37.78]	1.4212

The bootstrap uses 10,000 draws, seed 24092026, resampling five entire dossiers within each route with replacement. Each sampled dossier retains all its gates and all three trajectories. Percentile endpoints use linear interpolation. These intervals reflect only this small, stratified synthetic sample, not 255 independent observations or uncertainty over enterprises. In particular, zero observed all-three successes for DeepSeek is not proof of zero population success probability; its degenerate empirical bootstrap interval is not a useful risk bound.

Gemini has 255/255 correct dispositions, Luna 246/255, and DeepSeek 245/255. There are zero scored false approvals and critical misses for each model in this follow-up. Evidence fidelity fails on 10, 36, and 59 gates, respectively. These are new strict scores; the structural audit of the original 85 Gemini failures is not silently applied to these new traces.

The ranking persists in each repetition, but a similar overall rate does not imply repeatable success on every dossier. Complete-route pass/fail changes across repeats for five Gemini, six Luna, and eight DeepSeek dossiers. Only nine, three, and zero dossiers, respectively, pass all three times. Exact gate dispositions remain identical across repeats on all 15 Gemini dossiers, ten Luna dossiers, and six DeepSeek dossiers. Optional authorized conditional approvals can legitimately change a disposition, so disagreement is not by itself an error.

Provider routing was not fixed: DeepSeek records eight providers, Gemini Google AI Studio, and Luna OpenAI. Consequently, variability cannot be attributed solely to intrinsic model randomness. Temperature zero, small sample size, existing generated cases, and infrastructure resumption limit the interpretation to observed repeatability under this recorded protocol. All final scores, error checkpoints, costs, and supporting documents are released with the analysis script and per-repeat counts.

3.7 Dependence, uncertainty, and error propagation

The benchmark contains 300 underlying projects, not 5,094 independent enterprises. Multiple gates within a project share facts and handoffs; multiple models review the same project. The released case-cluster and paired analyses preserve those dependencies for the reported intervals. Those intervals still condition on the generated population and one retained trajectory per model and case. They do not measure uncertainty over organizations, model versions, or repeated inference runs. The follow-up in Section 3.6 separately measures three trajectories on 15 existing cases; it does not change the original intervals.

Sequential handoffs allow an upstream output to affect a later review. They make propagation observable in traces, but this single handoff condition does not identify a causal propagation effect. A controlled comparison would run the same cases with agent handoffs, reference handoffs, and no prior-review content, with matched resource budgets and repeated trajectories. Report downstream changes conditional on the upstream error class, rather than treating every repeated mention of one defect as an independent safety event.

The original gate definition is a useful strict endpoint; it is not the only endpoint an operating organization would choose. A deployment evaluation should separately register false approvals, missed critical findings, evidence validity, unauthorized actions, cycle time, and human work. Acceptance thresholds should follow the consequence of each error. The present study supplies observations for several of those outcomes, but no enterprise has validated its contract as a sufficient production acceptance test.

4 Discussion: where could a language model add value?

Once facts and executable policies are supplied, the rule engine achieves the measured endpoint without a language model. The useful comparison is therefore the cost and reliability of producing those inputs: extracting facts, resolving conflicting documents, translating policy, and obtaining missing information. Those costs are outside the original benchmark’s recorded API expenditure. A forward deployed engineer (FDE) would work with policy owners on these interfaces, exception handling, authority, and maintenance. These are deployment requirements, not measured productivity results.

An evidence interface should let the reviewer identify a source and a JSON path or document span, then attach the observed value deterministically. This avoids having a model reconstruct a literal

quotation. Evaluation should separately check source access, value fidelity, premise coverage, and inference. The structural sensitivity analysis evaluates one specified relaxation; it does not evaluate a deployed serializer or a semantic judge.

4.1 An executed preflight for document-only evaluation

Before treating removal of `REVIEW_FACTS` as an extraction benchmark, we tested whether remaining Word documents determine the reference decision. Using case `DGF-BUY-035004_buy`, we generated two offline variants differing only in the selected vendor’s due-diligence status: `complete` versus `pending`. The frozen evaluator requires `GO` in the first and `REWORK` with `PROC-DD-001` in the second. Yet all 26 generated Word documents have identical extracted paragraphs and table cells. The changed evidence is in the due-diligence and vendor-offer CSV records, and the procurement snapshot. The script, document text hashes, variant values, and reference outcomes are released. No model was called for this test.

This is an identifiability counterexample for a Word-only condition: the same input text can require different answers. It does not show that a document-plus-CSV condition is unsolvable, nor that document interpretation has no value. Additionally, the frozen strict scorer requires finding support to cite the gate’s `REVIEW_FACTS` identifier. Hiding that source while keeping this score would force evidence failures even when other material permits the correct decision. A fair ablation needs both an information-sufficiency check and a common revised evidence contract before paid evaluation.

The proposed factorial comparison crosses executable versus prose policy with structured snapshots versus source documents and operational records. Every cell must contain equivalent policy semantics and recoverable information, or explicitly score justified abstention on underdetermined cases. A rule-engine comparison should include the extraction stage and its engineering cost. A held-out set must be fixed independently of failures used to design these conditions. The released ablation protocol specifies these checks; paid ablation outcomes are not reported here.

A separate handoff intervention would replay the final General gate with either saved agent history or validated reference history, keeping its dossier, mandate, budget, and prompt identical otherwise. References must preserve the same verified upstream conditional approvals; blindly substituting unmodified labels would change both information and authority. Paired repeated runs would estimate the effect of handoff content on this protocol, rather than attributing every downstream failure to propagation.

4.2 Human work and the broader research hypothesis

Review automation could replace work currently performed by people if it reduces total human hours at comparable output, quality, and service. This requires measuring preparation, ordinary review, exceptions, correction, and recurring support, including suppliers. Initial integration and policy formalization must also be recorded for an investment comparison. Staffing, workload-equivalent full-time equivalents (FTE), and individual headcount are distinct outcomes. This benchmark measures none of them.

The companion manuscript develops a labor-accounting framework and an 80% FTE-reduction hypothesis for 2026–2033. That percentage is an author-selected research hypothesis, not an estimate supported by these scores. The separate prospective protocol uses seven years after a registered baseline ends; no cohort is enrolled, and that protocol cannot retrospectively confirm the original dated claim. These field questions motivate the project but are not empirical contributions of this benchmark study.

Task substitution and aggregate employment are distinct outcomes (Acemoglu and Restrepo, 2019). Residual oversight and maintenance matter even when routine execution succeeds (Bainbridge, 1983). Field productivity evidence such as Brynjolfsson et al. (2025) concerns observed work in an organization; its effect cannot be transferred to these synthetic project reviews.

5 Limitations and validation priorities

Task and external validity. The generator, public policy, and evaluator share an author-designed rule system. The comparator establishes sufficiency of this supplied system, not independent business validity. Decision balancing improves coverage but changes prevalence. There are 300 project fact sets and 267 whole-case finding patterns; some gate-level facts recur. The experiment does not measure discovery of tacit policy, real-world remediation, or human judgment. Independent specialists should validate rule meanings and adjudicate ambiguous cases.

Evidence metric. Exact excerpts test a serialization contract. The post-hoc lexical-or-structural endpoint is intentionally more permissive and was motivated by observed failures; it is not a new preregistered primary score or a semantic success rate. It retains original lexical passes, including overly broad quotations. Neither test establishes complete premise coverage. All-model diagnostics and unchanged primary scores are released so the effect of the relaxation is inspectable. A semantic evaluation requires independent labels for failed and passing evidence under a fixed rubric.

Sampling and repeated execution. The original study has one retained trajectory per model-case pair. The follow-up has only 15 existing dossiers and three trajectories per model; its cluster intervals condition on that sample. The three endpoints, including two Flash variants, are not representative of all systems. Provider routing was not fixed, and infrastructure resumes retain completed prefixes. No causal attribution of variability to intrinsic model randomness is made. Novel held-out cases, matched information conditions, and broader endpoint coverage are needed to test generalization.

Costs and baselines. API costs omit human formalization, document preparation, integration, local compute, maintenance, and review. The deterministic program has no model output-token limit and cites whole observed snapshots; its 100% result establishes task sufficiency, not an equal-resource optimization result. A qualified-human baseline and a fixed-budget comparison of complete extraction-and-decision systems remain absent.

Human and field validation. A future comparison should match information and authority across unaided humans, agents, and humans reviewing an agent draft; counterbalance cases and measure active work, correction, and downstream support. Review of a fluent draft may induce automation bias, so independent adjudication and error detection are necessary (Parasuraman and Manzey, 2010). There are no human participants or enterprise cohort in these data. Field quality, workload, and causal employment effects cannot be inferred from the benchmark.

The next paid studies should test policy representation, evidence extraction, and matched General-gate handoffs under a versioned protocol. The document-only preflight demonstrates why simply deleting supplied facts is insufficient. The original benchmark remains frozen; new information and evidence contracts must be reported as a separate experiment.

6 Conclusion

DGF-Bench measures rule application and evidence conformity under an explicit, favorable information condition. The models often make correct decisions, but a deterministic program solves the entire supplied contract. These results establish no incremental LLM advantage over that control. Comparing systems that must first extract facts or formalize policy is the next question, with the associated engineering and human costs included.

The all-model audit changes the interpretation of strict failures. Gemini’s scores are highly sensitive to excerpt serialization; the same relaxation removes substantially fewer DeepSeek and Luna failures. Complete confusion matrices, procurement diagnostics, and both initial and effective-reference agreement make the remaining errors inspectable. The 135-run follow-up preserves the ranking while showing that similar aggregate rates do not imply stable success for every dossier. A Word-only

counterexample demonstrates why an extraction ablation must first preserve enough information to determine the required answer.

These findings motivate an evidence interface that separates source attribution, premise coverage, rule application, and authorized decisions. They support testing how much review work such systems can perform in practice. They do not measure human work saved or a staffing effect; those hypotheses require operational observations under independent quality assessment.

Data, code, and research status

The public repository is <https://github.com/jeremy1392/dgf-agentic-bench>. The release **dgf-bench-300-20260923** contains the full recorded run (17,700 original files), prepared dataset (27,947 files), and benchmark source snapshot (1,099 files). It includes Word documents, architecture diagrams, simulated records, policies, traces, scores, usage ledgers, and historical failed attempts. Checksums and per-file inventories preserve provenance. Evaluator-only references are released for audit; they were not exposed by agent tools during the run.

The final data tables and reproduction instructions are in **research/2026-09-dgf-bench/**. Offline reproduction uses the frozen source archive, not an assumption that the evolving default branch is identical. The earlier 15-project pilot is separately available, including its Word documents, under **experiments/run_20260922_180730_063947/**; its outcomes are not pooled into this study. The second manuscript sources are in **paper2/**. Its inherited benchmark tables and figures retain the original measurements. The 300-case deterministic control and exhaustive all-model structural audit are released in **research/2026-09-followup/**. The follow-up also releases 135 completed model-case runs on 15 existing dossiers, their 765 scored gates, retained failed attempts, and USD 12.5598870228 in recorded costs. The new archive **dgf-bench-repetitions-20260924.zip** contains all 4,020 files retained in the repetition experiment directory, including the copied source dossiers and architecture documents. Decision confusion matrices and a reproducible document counterexample accompany these analyses. No paid scaffold ablation, human adjudication, or enterprise enrollment has been completed.

The author is Jeremy Canale, contact@jeremycanale.com, <https://www.jeremycanale.com>, <https://www.linkedin.com/in/jcanale13>. The workflow specification is author-designed; no confidential enterprise documents or employment outcomes are represented as observations. AI assistance was used to prepare and revise this manuscript and its supporting software; this is not independent expert adjudication. The author remains responsible for checking the claims, artifacts, and interpretation before submission.

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A Complete decision confusion matrices

Rows use the effective reference after validated authority actions; columns are the model’s submitted disposition. G = GO, C = GO_WITH_RESERVATIONS, R = REWORK, S = SUSPENSION, N = NO_GO. All original evaluable gates are included. The released summary also permits comparison with the separately reported initial-reference agreement.

Table 13: DeepSeek disposition confusion; 1700 evaluable gates.

Reference / predicted	G	C	R	S	N
G	441	3	35	9	0
C	0	302	23	1	0
R	0	0	416	2	1
S	0	0	3	186	2
N	1	0	1	0	274

Table 14: Gemini disposition confusion; 1694 evaluable gates.

Reference / predicted	G	C	R	S	N
G	488	0	0	0	0
C	0	398	0	0	0
R	0	0	352	0	0
S	0	0	0	180	0
N	0	0	0	0	276

Table 15: Luna disposition confusion; 1700 evaluable gates.

Reference / predicted	G	C	R	S	N
G	482	4	3	0	0
C	0	354	3	0	0
R	1	0	394	0	0
S	0	0	17	166	0
N	0	0	3	1	272

B Reproduction record

The public research package contains the complete original and repeated-run traces, unchanged primary scores, per-item all-model evidence classifications, strict and post-hoc route outcomes, confusion matrices, and counterfactual document hashes. Analysis scripts use the frozen benchmark source archive. They make no inference calls and do not overwrite original scores. The manuscript verification script checks published counts and the provenance of inherited tables and figures. These checks establish reproducibility and internal consistency; they do not provide independent professional validation of the rule system.